

Eco-Evolutionary Forecasting of Black-Browed Albatross

MATLAB code repository

This repository contains the MATLAB code used to generate the eco-evolutionary simulations and figures presented in the manuscript:

Climate mitigation contributes more to population persistence than evolutionary adaptation in a long-lived seabird

Jenouvrier, Garnier, Holland, van de Walle, Patrick, Bonnet, Delord, Ventura, Barbraud, and Weimerskirch.

MATLAB is proprietary software and requires a paid license. To improve accessibility, a general implementation of the EvoDemo-Hyper matrix population model has also been developed as an R package. The R package, together with documentation and examples, is available at <https://src.koda.cnrs.fr/timotheebonnet/evodemohypermpm.git> and is described in Van de Walle, J., Garnier, J., Bonnet, T., & Jenouvrier, S. (2025). *Toward a unified approach to modeling adaptation among demographers and evolutionary ecologists. Methods in Ecology and Evolution*.

The MATLAB code provided here extends this general hyperstate eco-evolutionary framework to the black-browed albatross population breeding at the Kerguelen Islands. The model integrates demographic state, breeding value, phenotype, and sea-surface-temperature forcing to project demographic and evolutionary dynamics under historical and future climate conditions.

The mathematical formulation and biological assumptions are described in the Supporting Information.

1. Model overview

The population is simultaneously classified by:

- demographic state i ;
- breeding-value class j ;
- phenotype class k .

The population vector at time t , denoted $\tilde{\mathbf{n}}_t$, contains the number of individuals in every combination of demographic state, breeding value, and phenotype.

Population dynamics follow:

$$\tilde{\mathbf{n}}_{t+1} = \tilde{\mathbf{A}}_{t,m} \tilde{\mathbf{n}}_t,$$

where m indexes an Earth-system-model ensemble member.

The projection matrix is decomposed into:

$$\tilde{\mathbf{A}}_{t,m} = \tilde{\mathbf{U}}_{t,m} + \tilde{\mathbf{F}}_{t,m},$$

where:

- $\tilde{\mathbf{U}}$ describes demographic and phenotypic transitions;
- $\tilde{\mathbf{F}}$ describes reproduction, inheritance, and offspring phenotype assignment.

The full hyperstate model is constructed from the following operators:

- $\mathbf{U}$: demographic survival and state transitions;
- $\mathbf{R}$: fertility and production of fledged female offspring;
- $\mathbf{B}$: breeding-value transitions within individuals;
- $\mathbf{P}$: phenotype transitions within individuals;
- $\mathbf{H}$: inheritance of offspring breeding values. In the MATLAB implementation, $\mathbf{H}$ is not assembled as an explicit matrix; inheritance is instead calculated directly using convolution and interpolation of the parental breeding-value distributions, followed by permute and reshape operations to restore the hyperstate ordering;
- $\mathbf{M}$: assignment of offspring phenotypes conditional on breeding value;
- permutation matrices used to reorder the dimensions of the hyperstate vector between sub-processes. In the MATLAB implementation, these permutations are performed implicitly through array indexing and reshaping operations rather than by constructing explicit permutation matrices.

The corresponding mathematical definitions are provided in the Supporting Information.

2. Demographic states

The demographic life cycle contains 13 states:

PB1, PB2, PB3, PB4, PB5, PB6, PB7, PB8, PB9+, S, F, NB

where:

- PB1–PB9+ are age-structured pre-breeders;
- S denotes successful breeders;
- F denotes failed breeders;
- NB denotes non-breeders that have bred previously.

Pre-breeders progress through age classes until recruitment. Recruitment can occur from age 5 onward. The final pre-breeder class is an open age class containing individuals aged 9 years or older that have not yet bred.

Once individuals recruit, they leave the age-structured pre-breeder states and enter one of the three adult breeding states. Adult breeders are not subsequently classified by age.

3. Quantitative-genetic structure

Under the infinitesimal model of quantitative genetics, phenotype is decomposed into additive genetic and environmental components:

$$V_P = V_A + V_E,$$

where:

- V_P is phenotypic variance;
- V_A is additive genetic variance;
- V_E is environmental variance.

Narrow-sense heritability is:

$$h^2 = \frac{V_A}{V_P}.$$

For a specified heritability and observed phenotypic variance:

$$V_A = h^2 V_P,$$

and

$$V_E = (1 - h^2) V_P.$$

The inheritance operator **H** assigns offspring breeding values using a Gaussian distribution centered on the mid-parent breeding value, with variance $V_A/2$.

The phenotype-assignment operator **M** assigns offspring phenotypes conditional on their breeding value using a Gaussian distribution with variance V_E .

The demographic model is female based. A 1:1 offspring sex ratio is assumed. For inheritance, the distribution of reproductive male breeding values is assumed to match that of reproductive females.

4. Phenotypes represented in the simulations

Each simulation follows one focal phenotype at a time. The variable PHENOTYPE selects the focal trait.

<i>PHENOTYPE</i>	<i>Trait</i>	<i>Trait category</i>
1	Activity: number of transitions between activity and rest on the water	Foraging behavior
2	Proportion of time spent on the water	Foraging behavior
3	Return date to the breeding colony	Phenological trait
4	Wing length measured at fledging	Morphological trait

The behavioral and phenological traits were measured using geolocator deployments. Wing length was measured at fledging.

Trait values used in the model are discretized into phenotype classes. The principal analyses use 100 phenotype classes.

5. Heritability scenarios

The principal simulations compare:

$$h^2 = 0.1$$
$$h^2 = 0.9$$

These values represent:

- low heritability and relatively weak additive genetic control;
- high heritability and an optimistic upper bound on adaptive potential.

Additional sensitivity analyses explore intermediate heritability values.

6. Environmental forcing

Annual environmental forcing is represented by the sea-surface-temperature vector:

$$\mathbf{E}_{t,m} = (E_{J,t,m}, E_{W,t,m}, E_{B,t,m}),$$

where:

- $E_{J,t,m}$ is SST in the juvenile wintering sector;
- $E_{W,t,m}$ is SST in the adult wintering sector;
- $E_{B,t,m}$ is SST in the adult breeding-season sector.

In the MATLAB parameter functions, these environmental variables are stored as:

```
SST(1,1) % Juvenile wintering-sector SST
SST(1,2) % Adult wintering-sector SST
SST(1,3) % Adult breeding-season SST
```

For each year t , climate ensemble member m , and phenotype class k , the vector of demographic parameters is:

$$\theta_{k,t,m} = \Theta(W_k' Z_k' E_{t,m}),$$

where:

- W_k is wing length;
- Z_k contains adult foraging activity, time on water, and return date;
- $E_{t,m}$ contains the three SST covariates.

The SST-vector definition and its links to demographic rates follow the Supporting Information notation.

7. Demographic parameter vector

The parameter functions return a $14 \times g$ matrix called theta, where g is the number of phenotype classes.

Each column corresponds to one phenotype class. The rows retain the following fixed order:

<i>Row</i>	<i>Parameter</i>
1	First-year survival
2	Pre-breeder survival
3	Survival of successful breeders
4	Survival of failed breeders and non-breeders
5	Recruitment probability at age 4
6	Recruitment probability at age 5
7	Recruitment probability at age 6
8	Recruitment probability at age 7
9	Recruitment probability at age 8 and older
10	Breeding success at first recruitment
11	Breeding probability of successful breeders
12	Breeding probability of failed breeders
13	Breeding probability of non-breeders
14	Adult breeding success

The demographic parameters depend on climate and phenotype as follows:

- first-year survival depends on juvenile-sector SST and wing length;
- recruitment depends on age and wing length;
- adult breeding probability depends on adult wintering-sector and breeding-season SST;
- breeding success at first recruitment depends on adult wintering-sector and breeding-season SST;
- adult breeding success depends on adult behavioral and phenological traits;
- adult survival is state specific but does not depend on climate or phenotype in the baseline parameterization.

These relationships are described in the Supporting Information and Jenouvrier et al. (2018).

8. Climate scenarios

Future projections use two Earth-system-model experiments.

High-emissions scenario

caseID = 1

This scenario follows SSP3-7.0 and uses the CESM2 Large Ensemble.

- model: CESM2;
- scenario: SSP3-7.0;
- ensemble size: 50 members;
- projection period: historical forcing through 2014 and future forcing from 2015 to 2100.

Low-emissions scenario

caseID = 3

This scenario represents a Paris-compatible 2°C stabilization pathway and uses the CESM1-CAM5 2.0°C never-exceed ensemble.

- model: CESM1-CAM5;
- scenario: 2.0°C never-exceed pathway;
- ensemble size: 11 members;
- ensemble members branch from the historical simulation in 2006.

Each climate ensemble member is treated as an equally plausible realization of climate evolution. Individual ensemble members are propagated through the eco-evolutionary model to represent internal climate variability. Ensemble means are used in analyses focused on the forced climate signal.

Files named `ClimateScenario_Case2.mat` and `ClimateScenario_Case4.mat` are included in the repository. The `ClimateScenario_Case*.mat` files are derived climate products rather than raw Earth-system-model outputs. They were generated from NetCDF files containing annual sea-surface temperature (SST) time series for the three biologically relevant regions used in the model: the juvenile wintering sector, the adult wintering sector, and the breeding-season sector. For the CESM2 Large Ensemble, regional SST time series were extracted from each ensemble member and assembled into MATLAB files for efficient use by the simulation code. The Paris-compatible 2°C scenario was processed in the same way from a dedicated NetCDF dataset.

All SST time series were converted to standardized anomalies using the 2006–2013 reference period. Anomalies were calculated using a common mean and standard deviation estimated across all climate ensemble members, allowing the environmental forcing to be expressed on a common standardized scale while preserving differences among ensemble realizations.

The climate forcing included in this repository is provided both as the original NetCDF files and as preprocessed MATLAB files used by the simulation code.

High-emissions scenario (SSP3–7.0, CESM2 Large Ensemble). The original climate forcing is distributed as four NetCDF files:

```
TEMP_timeseries_CESM2-LE_ens1.nc
TEMP_timeseries_CESM2-LE_ens2.nc
TEMP_timeseries_CESM2-LE_ens3.nc
TEMP_timeseries_CESM2-LE_ens4.nc
```

Each NetCDF file contains annual sea-surface temperature (SST) time series for **10 CESM2 Large Ensemble members**, yielding a total of **40 ensemble members**. For each ensemble member, three biologically relevant SST variables are provided:

- `ts_juvenile`: SST in the juvenile wintering sector,
- `ts_winter`: SST in the adult wintering sector,
- `ts_breeding`: SST in the breeding-season sector.

Low-emissions scenario (Paris-compatible 2°C stabilization). Climate forcing is provided in the NetCDF file:

```
SST_timeseries_Paris_2.nc
```

This file contains the corresponding annual SST time series for the same three oceanic regions across all ensemble members of the Paris-compatible climate simulation.

9. Historical and future parameterizations

Historical parameterization

```
parameterPHENOTYPE_HIST.m
```

This function defines the baseline climate–trait–vital-rate relationships used for historical simulations.

Future deterministic parameterization

parameterPHENOTYPE_ENV.m

This function uses the same mean demographic relationships as the historical parameterization. Environmental change among years and scenarios is introduced through the annual SST input.

Future stochastic parameterization

parameterPHENOTYPE_ENVSTO.m

This function adds stochastic variation to demographic rates around the mean climate–trait–demography relationships.

The stochastic deviations are intended to represent temporal demographic variability and uncertainty in the fitted vital-rate relationships. Annual SST from individual climate ensemble members and stochastic demographic deviations are propagated jointly through the population model.

Climate-dependent selection parameterization

parameterPHENOTYPE_ENV_SP.m

This function implements two juvenile-survival scenarios controlled by CaseSP.

CaseSP = 0

Baseline model: juvenile survival depends additively on juvenile-sector SST and wing length.

CaseSP = 1

Climate-dependent selection scenario: the effect of wing length on juvenile survival depends on SST, producing stronger selection on wing length under warmer conditions.

The corresponding scenarios are illustrated in Figure S2 and analyzed in Figures S3 and S4.

10. Sources of uncertainty

The projections account for two principal sources of uncertainty.

10.1 Internal climate variability

Each Earth-system-model ensemble member provides one possible trajectory of annual SST.

For each year and ensemble member, the SST vector is propagated through the demographic parameter functions to construct year-specific demographic matrices.

10.2 Demographic uncertainty and stochasticity

The stochastic parameter function adds Gaussian deviations to demographic relationships.

These deviations represent temporal variability in demographic rates and uncertainty in estimated climate–trait–demography relationships.

11. Parameter provenance

The climate–trait–demography relationships are based on the long-term demographic analysis of black-browed albatrosses at Kerguelen:

Jenouvrier, S., et al. (2018). Climate change and functional traits affect population dynamics of a long-lived seabird. *Journal of Animal Ecology*, 87, 906–920.

12. Repository structure

The repository is organized primarily by manuscript figure. Each figure folder contains:

1. a figure script;
2. a model-building and simulation script;
3. one or more parameter functions;
4. common model functions;
5. empirical or climate input data;
6. precomputed simulation outputs, where applicable;
7. the final figure file.

13. Figure 2: historical environment

Figure2/

This folder contains simulations under the historical parameterization.

Figure-specific files

MainFig2.m
main_Pheno_ENV_HIST.m
parameterPHENOTYPE_HIST.m
figure2.pdf
resultsMATfiles.zip

MainFig2.m

Loads the relevant simulation outputs and generates Figure 2.

main_Pheno_ENV_HIST.m

Builds and runs the hyperstate eco-evolutionary model under historical environmental conditions.

parameterPHENOTYPE_HIST.m

Calculates the 14 demographic parameters under the historical parameterization.

resultsMATfiles.zip

Contains precomputed simulation results used to reproduce the figure.

The README should be updated with a precise statement indicating whether historical SST is held at a fixed historical value, represented by a historical mean, or supplied as an observed annual time series. The historical climate is represented by a constant vector of observed mean SST anomalies, calculated from the 1986–2022 observational record for the juvenile wintering, adult wintering, and breeding sectors. Thus, historical simulations assume constant environmental conditions equal to the long-term observed average rather than reproducing the observed year-to-year SST variability.

```
matlab
SST = [-0.7563 -0.3128 -0.8093];
```

14. Figure 3: future climate projections with uncertainty

Figure3/

This folder contains projections under future climate change with uncertainty from both climate variability and demographic processes.

Figure-specific files

MainFig3.m
main_Pheno_ENV_ClimateChange_UNC.m
parameterPHENOTYPE_ENV.m
parameterPHENOTYPE_ENVSTO.m
ClimateScenario_Case1.mat
ClimateScenario_Case2.mat
ClimateScenario_Case3.mat
ClimateScenario_Case4.mat

MainFig3.m

Loads ensemble simulation outputs and generates Figure 3.

main_Pheno_ENV_ClimateChange_UNC.m

Builds and runs future eco-evolutionary projections while propagating climate and demographic uncertainty.

parameterPHENOTYPE_ENV.m

Calculates the mean climate- and phenotype-dependent demographic rates.

parameterPHENOTYPE_ENVSTO.m

Adds stochastic deviations to demographic-rate relationships.

ClimateScenario_Case*.mat

Contains annual SST forcing for the climate scenarios and ensemble members.

ClimateScenario_Case1.mat. This file contains standardized annual sea-surface temperature (SST) anomalies for the historical-to-future period (2006–2100) under the high-emissions CESM2 Large Ensemble scenario. Three variables are provided: `ts_juvenilea`, `ts_wintera`, and `ts_breedinga`, corresponding to the juvenile wintering, adult wintering, and breeding sectors, respectively. Each variable is stored as a **95 × 40** matrix of double-precision values, where rows correspond to years (2006–2100) and columns correspond to the **40 climate ensemble members**. SST values are standardized anomalies (dimensionless), obtained by subtracting the common mean and dividing by the common standard deviation calculated across all ensemble members over the 2006–2013 reference period.

`ClimateScenario_Case2.mat`, `ClimateScenario_Case3.mat`, and `ClimateScenario_Case4.mat` have the same variables, dimensions, units, and organization as `ClimateScenario_Case1.mat`, differing only in the climate scenario represented by the corresponding simulation case.

15. Figure 4: climate-dependent selection

Figure4/

This folder explores an alternative selective-pressure scenario affecting wing length and juvenile survival.

Figure-specific files

MainFig4.m

main_Pheno_ENV_ClimateChange_meanEns.m

parameterPHENOTYPE_ENV_SP.m

figure4.pdf

ecoEvoResults_*.mat

MainFig4.m

Generates Figure 4.

main_Pheno_ENV_ClimateChange_meanEns.m

Runs the eco-evolutionary model using climate ensemble means under baseline and climate-dependent selection scenarios.

parameterPHENOTYPE_ENV_SP.m

Calculates demographic parameters under the two juvenile-selection scenarios controlled by CaseSP.

ecoEvoResults_*.mat

Contains precomputed simulation outputs used to generate the figure.

16. Common model functions

The following functions are shared across figure folders.

Umat2.m

Fmat2.m

invlogit.m

invlogitG.m

Umat2.m

Constructs the 13×13 demographic survival and stage-transition matrix for a given phenotype class, breeding-value class, year, and climate ensemble member.

It combines:

- juvenile survival;
- pre-breeder survival;
- recruitment;
- adult survival;
- adult breeding probability;
- breeding success.

Fmat2.m

Constructs the 13×13 reproduction matrix.

Only the first row contains nonzero fertility entries because offspring enter the first pre-breeder state.

Fertility incorporates:

- parental survival;
- breeding or recruitment probability;
- breeding success;
- an assumed female offspring fraction of 1/2.

The demographic definitions of the transition and reproduction matrices are given in the Supporting Information.

invlogit.m

Transforms linear predictors into probabilities using the inverse-logit function.

invlogitG.m

Provides an inverse-logit transformation used by selected model components.

17. Empirical input files

FOR.mat
wing_corrected.mat
Nobs_FRA.mat

FOR.mat Contains empirical measurements of behavioral and phenological traits.

The expected trait order is:

Column 1: foraging activity
Column 2: proportion of time spent on the water
Column 3: return date

wing_corrected.mat Contains wing-length measurements at fledging.

Nobs_FRA.mat contains the observed annual population size of the black-browed albatross population breeding at the Kerguelen Islands. These observations are used for model initialization, hindcasting, and validation of population projections. The dataset is derived from long-term demographic monitoring of the Kerguelen population described in Ventura et al. (submitted).

18. Program architecture

Each principal simulation script follows three broad blocks.

Block 1: parameter initialization

- load trait, demographic, and climate data;
- select the focal phenotype;
- define heritability;
- select the climate scenario;
- select the selective-pressure scenario when applicable;
- define the discretized phenotype and breeding-value classes;
- initialize the demographic state distribution.

Block 2: model construction and simulation

- calculate V_A and V_E ;
- construct demographic matrices $\mathbf{U}$ and $\mathbf{R}$;
- construct breeding-value operator $\mathbf{B}$;
- construct phenotype-transition operator $\mathbf{P}$;
- construct inheritance operator $\mathbf{H}$;
- construct phenotype-assignment operator $\mathbf{M}$;
- assemble the hyperstate transition and reproduction matrices;
- run the burn-in period;
- run the forward projection;
- save outputs for each phenotype, heritability, climate scenario, ensemble member, and selective-pressure scenario.

Block 3: derived outputs and figures

The scripts calculate:

- total population size;
- annual population growth;
- mean phenotype;
- mean breeding value;
- phenotypic variance;
- additive genetic variance;
- demographic-rate trajectories;
- adaptation rates;
- final phenotype and breeding-value distributions;
- uncertainty envelopes.

These outputs are then used by the MainFigX.m scripts.

19. Running the code

A typical workflow is:

cd Figure2

MainFig2

or:

cd Figure3
MainFig3

or:

cd Figure4
MainFig4

Before running a figure script:

1. ensure that all .mat input files are present;
2. unzip precomputed result archives where required;
3. add the figure folder and shared function folders to the MATLAB path;
4. confirm that phenotype, heritability, climate case, and scenario identifiers are set correctly;
5. confirm whether the script runs new simulations or loads precomputed outputs.

20. Citation

When using this code, please cite the accompanying manuscript and the methodological references describing the demographic and hyperstate models.

Manuscript

Jenouvrier, S., Garnier, J., Holland, M., van de Walle, J., Patrick, S., Bonnet, T., Delord, K., Ventura, F., Barbraud, C., and Weimerskirch, H.

Climate mitigation contributes more to population persistence than evolutionary adaptation in a long-lived seabird.

Demographic parameterization

Jenouvrier, S., et al. (2018). Climate change and functional traits affect population dynamics of a long-lived seabird. *Journal of Animal Ecology*, 87, 906–920.

Hyperstate eco-evolutionary model

Van de Walle, J., Garnier, J., Bonnet, T., and Jenouvrier, S. (2025). Toward a unified approach to modeling adaptation among demographers and evolutionary ecologists. *Methods in Ecology and Evolution*.

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